LAB 6- Anomaly detection in GKE with Prophet and Istio.

Lab Task 1 – Train a Prophet model with seasonality for normal operation

I provisioned my cluster using the provided provisioned script with the name as shown.

```
oadewusi@oluSec:~/aiops/Assignments/LAB6$ ./provision2.sh onlineboutique aiops4 europe-west4

 No current clusters found, continuing to deploy one
 Note: The Kubelet readonly port (10255) is now deprecated. Please update your workloads to use the recommended alternatives. Se
 readonly-port for ways to check usage and for migration instructions.
 Note: Your Pod address range (`--cluster-ipv4-cidr`) can accommodate at most 1008 node(s).
 Creating cluster onlineboutique in europe-west4-a... Cluster is being health-checked (Kubernetes Control Plane is healthy)...do
 Created [https://container.googleapis.com/v1/projects/aiops4/zones/europe-west4-a/clusters/onlineboutique].
 To inspect the contents of your cluster, go to: https://console.cloud.google.com/kubernetes/workload_/gcloud/europe-west4-a/onl
 kubeconfig entry generated for onlineboutique.
 NAME            LOCATION        MASTER_VERSION       MASTER_IP     MACHINE_TYPE   NODE_VERSION         NUM_NODES  STATUS
 onlineboutique  europe-west4-a  1.30.5-gke.1443001  34.91.38.113  e2-standard-4  1.30.5-gke.1443001  1          RUNNING
```

The went ahead to scale the replicas of loadgenerator down to 0.

```
oadewusi@oluSec:~/aiops/Assignments/LAB6$ kubectl scale deploy loadgenerator --replicas=0
 deployment.apps/loadgenerator scaled
oadewusi@oluSec:~/aiops/Assignments/LAB6$ kubectl get deploy
 NAME                    READY   UP-TO-DATE   AVAILABLE   AGE
 adservice               1/1     1            1           98s
 cartservice             1/1     1            1           97s
 checkoutservice         1/1     1            1           97s
 currencyservice         1/1     1            1           96s
 emailservice            1/1     1            1           96s
 frontend                1/1     1            1           96s
 istio-gateway-istio     1/1     1            1           93s
 loadgenerator           0/0     0            0           95s
 paymentservice          1/1     1            1           95s
 productcatalogservice   1/1     1            1           95s
 recommendationservice   1/1     1            1           94s
 redis-cart              1/1     1            1           94s
 shippingservice         1/1     1            1           94s
```

Got the IP address of the istio-gateway-istio – LoadBalancer – 34.13.169.74

```
oadewusi@oluSec:~/aiops/Assignments/LAB6$ kubectl get svc
 NAME                    TYPE           CLUSTER-IP       EXTERNAL-IP    PORT(S)                        AGE
 adservice               ClusterIP      34.118.237.46    <none>         9555/TCP                       26h
 cartservice             ClusterIP      34.118.232.174   <none>         7070/TCP                       26h
 checkoutservice         ClusterIP      34.118.227.140   <none>         5050/TCP                       26h
 currencyservice         ClusterIP      34.118.236.41    <none>         7000/TCP                       26h
 emailservice            ClusterIP      34.118.230.64    <none>         5000/TCP                       26h
 frontend                ClusterIP      34.118.226.175   <none>         80/TCP                         26h
 istio-gateway-istio     LoadBalancer   34.118.227.187   34.13.169.74   15021:31358/TCP,80:30530/TCP   26h
 kubernetes              ClusterIP      34.118.224.1     <none>         443/TCP                        26h
 paymentservice          ClusterIP      34.118.237.163   <none>         50051/TCP                      26h
 productcatalogservice   ClusterIP      34.118.231.178   <none>         3550/TCP                       26h
 recommendationservice   ClusterIP      34.118.232.210   <none>         8080/TCP                       26h
 redis-cart              ClusterIP      34.118.234.230   <none>         6379/TCP                       26h
 shippingservice         ClusterIP      34.118.228.225   <none>         50051/TCP                      26h
oadewusi@oluSec:~/aiops/Assignments/LAB6$
```

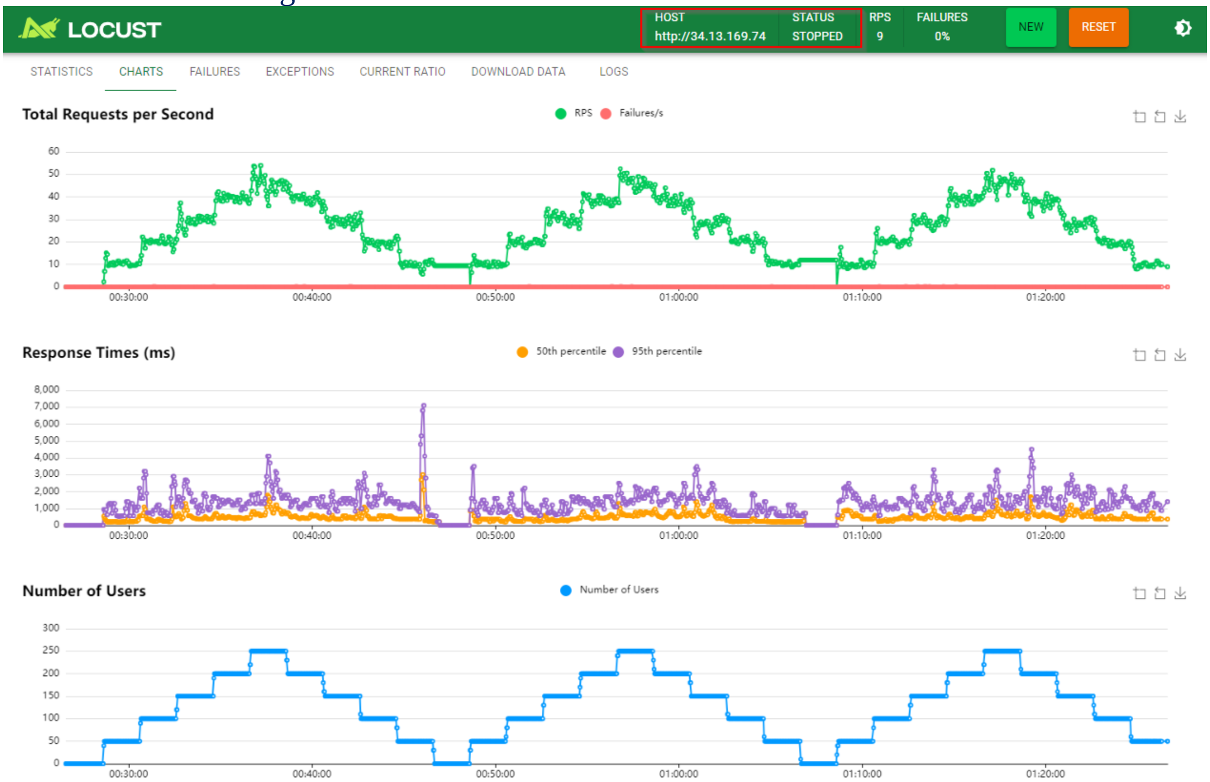
Then started the locust load generator with locust --host=’’http://34.13..169.74” -f locustfile_step_new.py

```
oadewusi@oluSec:~/aiops/Assignments/LAB6$ locust --host="http://34.13.169.74" -f locustfile_step_new.py
 Test run time is  3600
 Spawn rate is  10
 Cycle time is  1200
 Users is  250
 Number of steps is  10
 Seconds per step is  120
 Users per step is  50
 [2024-11-24 00:25:50,108] oluSec/INFO/locust.main: Starting Locust 2.32.2
 [2024-11-24 00:25:50,109] oluSec/INFO/locust.main: Starting web interface at http://0.0.0.0:8089
```

The locust ran for 3600s and stopped.

```
last_target_users = 50
run time = 3598.751143837002
current users = 50
last_target_users = 50
run time = 3599.7524928720013
current users = 50
last_target_users = 50
run time = 3600.753331991
current users = 50
last_target_users = 50
[2024-11-24 01:26:33,376] oluSec/INFO/locust.runners: Shape test stopping
```

The locust UI showing the charts.



Then edited the Prometheus to be a LoadBalancer to get an external IP: 34.34.58.54

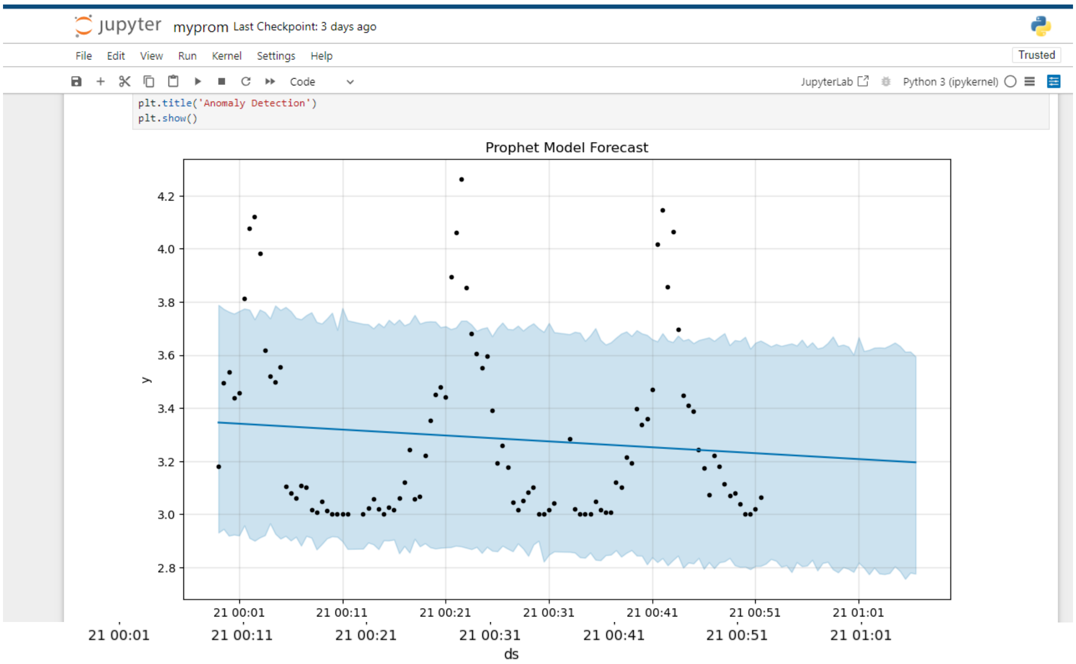
```
oadewusi@oluSec:~/aiops/Assignments/LAB6$ kubectl edit svc prometheus -n istio-system
service/prometheus edited
oadewusi@oluSec:~/aiops/Assignments/LAB6$ kubectl get svc -n istio-system
NAME                   TYPE           CLUSTER-IP       EXTERNAL-IP     PORT(S)                                                                       AGE
grafana                ClusterIP      34.118.228.253   <none>          3000/TCP                                                                      128m
istio-egressgateway    ClusterIP      34.118.226.227   <none>          80/TCP,443/TCP                                                                129m
istio-ingressgateway   LoadBalancer   34.118.228.117   34.32.195.246   15021:31040/TCP,80:31878/TCP,443:31804/TCP,31400:30657/TCP,15443:31273/TCP   129m
istiod                 ClusterIP      34.118.233.113   <none>          15010/TCP,15012/TCP,443/TCP,15014/TCP                                         129m
jaeger-collector       ClusterIP      34.118.233.120   <none>          14268/TCP,14250/TCP,9411/TCP,4317/TCP,4318/TCP                                128m
kiali                  ClusterIP      34.118.239.123   <none>          20001/TCP,9090/TCP                                                            128m
loki                   ClusterIP      34.118.239.154   <none>          3100/TCP,9095/TCP                                                             128m
loki-headless          ClusterIP      None             <none>          3100/TCP                                                                      128m
loki-memberlist        ClusterIP      None             <none>          7946/TCP                                                                      128m
prometheus             LoadBalancer   34.118.226.1     34.34.58.54     9090:31518/TCP                                                                128m
tracing                ClusterIP      34.118.239.196   <none>          80/TCP,16685/TCP                                                              128m
zipkin                 ClusterIP      34.118.227.244   <none>          9411/TCP                                                                      128m
```

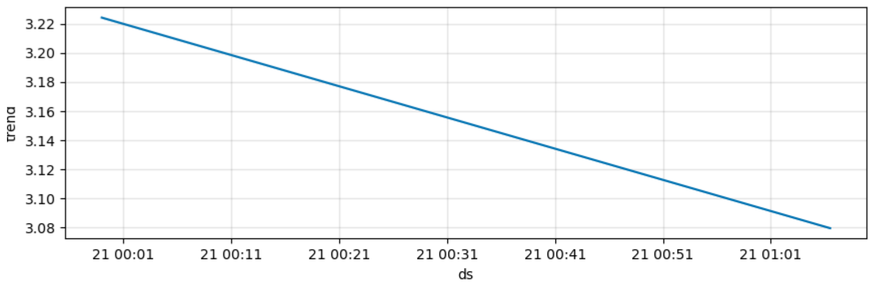
Used this curl query to get the training data sending it to onlineboutique.json

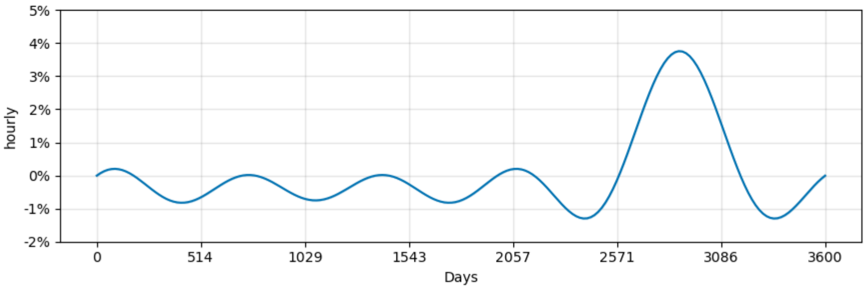
curl -g -s "http://34.34.58.54:9090/api/v1/query" --data-urlencode "query=histogram_quantile(0.5, rate(istio_request_duration_milliseconds_bucket{source_app='frontend', destination_app='shippingservice', reporter='source'}[1m]))[40m:30s]" | jq | more > onlineboutique.json

```
oadewusi@oluSec:~/aiops/Assignments/LAB6$ curl -g -s "http://34.34.58.54:9090/api/v1/query" --data-urlencode "query=histogram_quantile( 0.5, rate(istio_request_duration_milliseconds_bucket{source_app='frontend', destination_app='shippingservice', reporter='source'}[1m]))[40m:30s]" | jq | more > onlineboutique.json
```

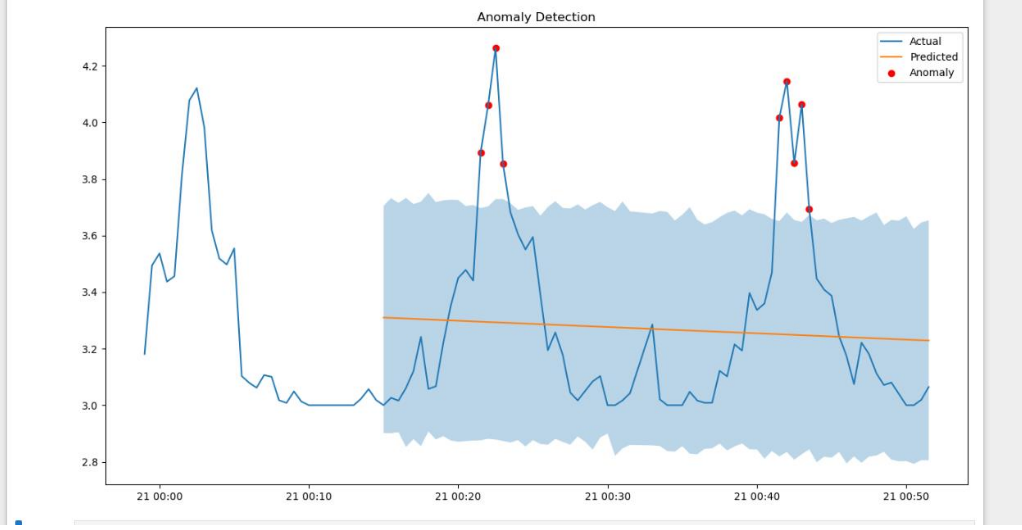
Screenshot of myprom.ipynb graphs (notebook uploaded to GitHub)







Days



Lab Task 2 – Verify no/minimal anomaly detection under normal operation

Implemented a monitor.py application.

```
monitor.py ×
monitorapp > monitor.py
import time
import requests
import pandas as pd
import numpy as np
from prophet import Prophet
from datetime import datetime, timedelta
from prometheus_client import start_http_server, Gauge
from sklearn.metrics import mean_absolute_error
import json

# Define Prometheus gauges
anomaly_gauge = Gauge('anomaly_count', 'Number of anomalies detected')
mae_gauge = Gauge('mae_score', 'Mean Absolute Error')
mape_gauge = Gauge('mape_score', 'Mean Absolute Percentage Error')
y_min_gauge = Gauge('y_min', 'Lower bound of prediction')
y_gauge = Gauge('y', 'Actual value')
y_max_gauge = Gauge('y_max', 'Upper bound of prediction')

# Prometheus URL and query
url = 'http://prometheus.istio-system:9090/api/v1/query'
query = 'histogram_quantile(0.5, rate(istio_request_duration_milliseconds_bucket{source_app="frontend", destination_app="shippingservice", reporter=

def load_training_data():
    with open('boutique_train.json', 'r') as file:
        data = json.load(file)

    values = data['data']['result'][0]['values']
    df_train = pd.DataFrame(values, columns=['ds', 'y'])

    df_train['ds'] = pd.to_datetime(df_train['ds'], unit='s')
    df_train['y'] = df_train['y'].astype(float)

    df_train['ds'] = df_train['ds'] - df_train['ds'].iloc[0]
    df_train['ds'] = df_train['ds'].dt.total_seconds()
    df_train['ds'] = df_train['ds'].apply(lambda sec: datetime.fromtimestamp(sec))

    return df_train
```

Display of monitor.py application indicating minimal difference between the **observed** and the **predicted** value which also gives low MAPE and MAE score with minimal anomaly observed.

```
oadewusi@oluSec:~/aiops/Assignments/LAB6$ python3 monitor.py
Importing plotly failed. Interactive plots will not work.
13:31:44 - cmdstanpy - INFO - Chain [1] start processing
13:31:44 - cmdstanpy - INFO - Chain [1] done processing

  ID          Timestamp          Anomalies    Observed      Predicted       MAE          MAPE
 -----------------------------------------------------------------------------------------------
  0       2024-11-24 13:31:45       0         0.000000      0.000000      0.000000      0.000000
  1       2024-11-24 13:32:45       0         0.000000      0.000000      0.000000      0.000000
  2       2024-11-24 13:33:45       0         0.000000      0.000000      0.000000      0.000000
  3       2024-11-24 13:34:46       0         3.000000      3.578066      0.578066      0.192689
  4       2024-11-24 13:35:46       0         3.052980      3.508605      0.455625      0.149239
  5       2024-11-24 13:36:47       0         3.101266      3.448053      0.346788      0.111821
  6       2024-11-24 13:37:47       0         3.008850      3.397751      0.388902      0.129253
  7       2024-11-24 13:38:47       0         3.229885      3.355986      0.126101      0.039042
  8       2024-11-24 13:39:48       0         3.425676      3.321611      0.104065      0.030378
  9       2024-11-24 13:40:48       1         4.223684      3.295094      0.928591      0.219853
  10      2024-11-24 13:41:48       1         4.163880      3.275153      0.888727      0.213437
  11      2024-11-24 13:42:49       1         6.157588      3.260944      2.896644      0.470419
  12      2024-11-24 13:43:49       1         6.098131      3.252261      2.845870      0.466679
  13      2024-11-24 13:44:50       0         3.637427      3.248128      0.389298      0.107026
  14      2024-11-24 13:45:50       1         4.133333      3.248005      0.885329      0.214192
  15      2024-11-24 13:46:50       0         3.189112      3.251117      0.062005      0.019443
  16      2024-11-24 13:47:51       0         3.237082      3.256908      0.019826      0.006125
  17      2024-11-24 13:48:51       0         3.023166      3.264506      0.241340      0.079830
  18      2024-11-24 13:49:51       0         3.175258      3.273314      0.098056      0.030881
  19      2024-11-24 13:50:52       0         3.036364      3.282814      0.246451      0.081166
  20      2024-11-24 13:51:52       0         3.027027      3.292013      0.264986      0.087540
  21      2024-11-24 13:52:53       0         0.000000      0.000000      0.000000      0.000000
  22      2024-11-24 13:53:53       0         3.000000      3.307370      0.307370      0.102457
```

Lab Task 3 – Inject faults with Istio to detect anomalies

delay_injection.yaml file.

```yaml
apiVersion: networking.istio.io/v1beta1
kind: VirtualService
metadata:
  name: shippingservice
spec:
  hosts:
  - shippingservice
  http:
  - match:
    - sourceLabels:
        app: "frontend"
    fault:
      delay:
        fixedDelay: 2s
        percent: 100
    route:
    - destination:
        host: shippingservice
  - route:
    - destination:
        host: shippingservice
```

Display of console output:
Marked yellow space – indicating before fault introduction.
Marked red space - indicating when fault was introduced.
Marked blue space – indicating when fault was removed.

```
oadewusi@oluSec:~/aiops/Assignments/LAB6$ python3 monitor.py
Importing plotly failed. Interactive plots will not work.
13:04:12 - cmdstanpy - INFO - Chain [1] start processing
13:04:12 - cmdstanpy - INFO - Chain [1] done processing

ID   Timestamp             Anomalies   Observed      Predicted   MAE           MAPE
--------------------------------------------------------------------------------------
0    2024-11-24 13:04:13   0           0.000000      0.000000    0.000000      0.000000
1    2024-11-24 13:05:13   0           0.000000      0.000000    0.000000      0.000000
2    2024-11-24 13:06:13   0           3.000000      3.659404    0.659404      0.219801
3    2024-11-24 13:07:14   0           3.018018      3.578066    0.560048      0.185568
4    2024-11-24 13:08:14   0           3.027586      3.508605    0.481019      0.158879
5    2024-11-24 13:09:15   0           3.103286      3.448053    0.344767      0.111097
6    2024-11-24 13:10:15   0           3.100840      3.397751    0.296911      0.095752
7    2024-11-24 13:11:15   1           1750.000000   3.355986    1746.644014   0.998082
8    2024-11-24 13:12:16   1           1750.000000   3.321611    1746.678389   0.998102
9    2024-11-24 13:13:16   1           1750.000000   3.295094    1746.704906   0.998117
10   2024-11-24 13:14:17   1           1750.000000   3.274872    1746.725128   0.998129
11   2024-11-24 13:15:17   1           1750.000000   3.260944    1746.739056   0.998137
12   2024-11-24 13:16:17   1           1750.000000   3.252261    1746.747739   0.998142
13   2024-11-24 13:17:18   0           3.569444      3.248128    0.321316      0.090018
14   2024-11-24 13:18:18   0           3.623457      3.248005    0.375452      0.103617
15   2024-11-24 13:19:19   0           3.357895      3.251117    0.106778      0.031799
16   2024-11-24 13:20:19   0           3.155763      3.256908    0.101145      0.032051
17   2024-11-24 13:21:19   0           3.067511      3.264506    0.196995      0.064220
18   2024-11-24 13:22:20   0           3.104762      3.273467    0.168705      0.054337
```

Lab Task 4 – Deploy your monitor into the Boutique cluster and publish to Grafana.

Dockerfile:

```
Dockerfile ×
monitorapp > Dockerfile > ...
FROM python:3.9-slim
RUN apt-get update && apt-get install -y \
    build-essential \
    python3-dev \
    gcc \
    && rm -rf /var/lib/apt/lists/*
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY monitor.py .
COPY boutique_train.json .
EXPOSE 8000
ENV PYTHONUNBUFFERED=1
CMD ["python", "monitor.py"]
```

Building the docker image

```
oadewusi@oluSec:~/aiops/Assignments/LAB6$ cd monitorapp/
oadewusi@oluSec:~/aiops/Assignments/LAB6/monitorapp$ docker build -t oadewusi/monitorapp:latest .
[+] Building 3.6s (13/13) FINISHED                                          docker:default
 => [internal] load build definition from Dockerfile                                  0.0s
 => => transferring dockerfile: 384B                                                  0.0s
 => [internal] load metadata for docker.io/library/python:3.9-slim                    3.1s
 => [auth] library/python:pull token for registry-1.docker.io                         0.0s
 => [internal] load .dockerignore                                                     0.0s
 => => transferring context: 2B                                                       0.0s
 => [internal] load build context                                                     0.0s
```

Docker push to the Docker hub:

```
oadewusi@oluSec:~/aiops/Assignments/LAB6/monitorapp$ docker tag oadewusi/monitorapp oadewusi/monitorappv4
oadewusi@oluSec:~/aiops/Assignments/LAB6/monitorapp$ docker push oadewusi/monitorappv4
Using default tag: latest
The push refers to repository [docker.io/oadewusi/monitorappv4]
4a7ca49069f1: Pushed
f2739d47f649: Pushed
0ce17a81f30a: Mounted from oadewusi/monitorappv3
51648f3f8291: Mounted from oadewusi/monitorappv3
c3e483e788a8: Mounted from oadewusi/monitorappv3
7ccf20d8cefa: Mounted from oadewusi/monitorappv3
aacba17e24d9: Mounted from oadewusi/monitorappv3
f751ad7c65c4: Mounted from oadewusi/monitorappv3
7822e749b484: Mounted from oadewusi/monitorappv3
c3548211b826: Mounted from oadewusi/monitorappv3
latest: digest: sha256:8c4d29379686b62f15e09ab7cbe707b880f34692678b40b10ab80cd70cd1f452 size: 2413
```

monskeleton_deploy.yaml file.

```
! monskeleton_deploy.yaml ×

! monskeleton_deploy.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: monitorappdeploy
  labels:
    app: monitor
spec:
  replicas: 0
  selector:
    matchLabels:
      app: monitorapp
  template:
    metadata:
      labels:
        app: monitorapp
      annotations:
        prometheus.io/scrape: "true"
        prometheus.io/port: "8000"
        prometheus.io/path: "/metrics"
    spec:
      containers:
      - name: moncontainer
        imagePullPolicy: Always
        image: index.docker.io/oadewusi/monitorappv4:latest
        ports:
        - containerPort: 8000
        resources:
          requests:
            cpu: 200m
```

Applying the monitorapp to the Kubernetes cluster and scaling the monitorappdeploy up to 1 replica

```
oadewusi@oluSec:~/aiops/Assignments/LAB6$ kubectl apply -f monskeleton_deploy.yaml
deployment.apps/monitorappdeploy created
oadewusi@oluSec:~/aiops/Assignments/LAB6$ kubectl scale deploy monitorappdeploy --replicas=1
deployment.apps/monitorappdeploy scaled
oadewusi@oluSec:~/aiops/Assignments/LAB6$ kubectl get pods
NAME                                     READY   STATUS    RESTARTS       AGE
adservice-64586ccfb9-zjndk               2/2     Running   0              25h
cartservice-6bd5c944b4-9cqxh             2/2     Running   0              25h
checkoutservice-6d4bcd6f95-gsg82         2/2     Running   0              25h
currencyservice-f987888c-mtd7m           2/2     Running   5 (7m5s ago)   25h
emailservice-56fbcfbbf7-cr9pr            2/2     Running   0              25h
frontend-cb9967686-4lbch                 2/2     Running   0              25h
istio-gateway-istio-5b8576f55b-zxz9x     1/1     Running   0              25h
monitorappdeploy-7685745dbc-jh4r2        2/2     Running   0              17s
paymentservice-794b5dfdf7-g79z6          2/2     Running   4 (167m ago)   25h
productcatalogservice-74b4c878d5-8gcfn   2/2     Running   0              25h
recommendationservice-65dd5bd87c-29rhq   2/2     Running   0              25h
redis-cart-7ff8f4d6ff-lwqkb              2/2     Running   0              25h
shippingservice-699bcb7fd5-flrt7         2/2     Running   0              25h
```

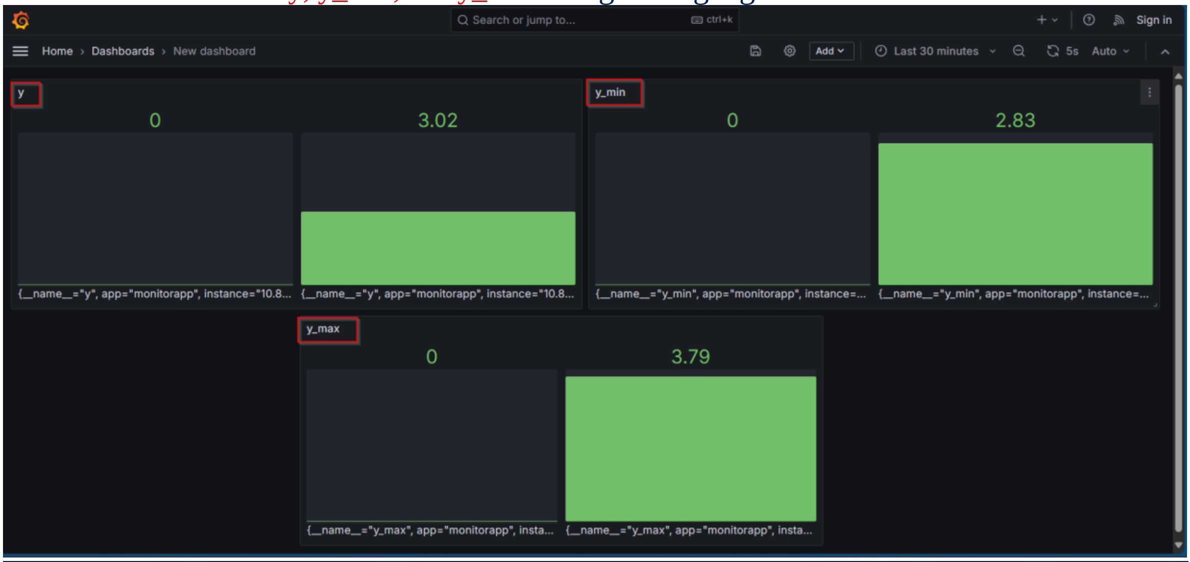
Using the command below to check the operation of the monitor app while running the locust generator.

```
oadewusi@oluSec:~/aiops/Assignments/LAB6$ kubectl logs -f monitorappdeploy-7685745dbc-jh4r2
Importing plotly failed. Interactive plots will not work.
20:50:53 - cmdstanpy - INFO - Chain [1] start processing
20:50:53 - cmdstanpy - INFO - Chain [1] done processing

 ID        Timestamp          Anomalies   Observed    Predicted     MAE        MAPE
-----------------------------------------------------------------------------------------
 0    2024-11-24 20:50:53        0       0.000000    0.000000    0.000000    0.000000
 1    2024-11-24 20:51:54        0       0.000000    0.000000    0.000000    0.000000
 2    2024-11-24 20:52:54        0       0.000000    0.000000    0.000000    0.000000
 3    2024-11-24 20:53:54        0       3.000000    3.579353    0.579353    0.193118
 4    2024-11-24 20:54:54        0       3.000000    3.509722    0.509722    0.169907
 5    2024-11-24 20:55:54        0       3.097902    3.449925    0.352023    0.113633
 6    2024-11-24 20:56:54        0       3.062257    3.399326    0.337070    0.110072
 7    2024-11-24 20:57:54        0       3.152000    3.357286    0.205286    0.065129
 8    2024-11-24 20:58:54        0       3.313993    3.323158    0.009165    0.002766
 9    2024-11-24 20:59:55        0       3.361290    3.295904    0.065387    0.019453
 10   2024-11-24 21:00:55        1       4.250814    3.275753    0.975061    0.229382
 11   2024-11-24 21:01:55        0       3.996845    3.261543    0.735302    0.183971
 12   2024-11-24 21:02:55        1       6.401099    3.252613    3.148486    0.491866
 13   2024-11-24 21:03:55        1       5.765306    3.248299    2.517007    0.436578
 14   2024-11-24 21:04:55        1       4.095082    3.247935    0.847147    0.206869
 15   2024-11-24 21:05:56        0       4.098305    3.250924    0.847381    0.206764
 16   2024-11-24 21:06:56        0       3.221519    3.256488    0.034969    0.010855
 17   2024-11-24 21:07:56        0       3.199357    3.263980    0.064624    0.020199
 18   2024-11-24 21:08:56        0       3.082645    3.272727    0.190083    0.061662
 19   2024-11-24 21:09:56        0       3.060870    3.282055    0.221186    0.072262
 20   2024-11-24 21:10:56        0       3.000000    3.291291    0.291291    0.097097
```

Grafana dashboard of the Request Response time using : histogram_quantile(0.5, rate(istio_request_duration_milliseconds_bucket{source_app="frontend", destination_app="shippingservice", reporter="source"}[1m]))

Grafana dashboard of y, y_min, and y_max using a bar gauge



Grafana dashboard of y, y_min, and y_max using a timeseries

Grafana dashboard of anomaly_count, mae_score, and mape_score using a timeseries for last 15 minutes

Grafana dashboard of anomaly_count, mae_score, and mape_score using a timeseries for last 30 minutes